EES 102: Introduction to Environmental Sciences

Instructors – Prof. Baerbel Sinha and Dr. Sunil A. Patil

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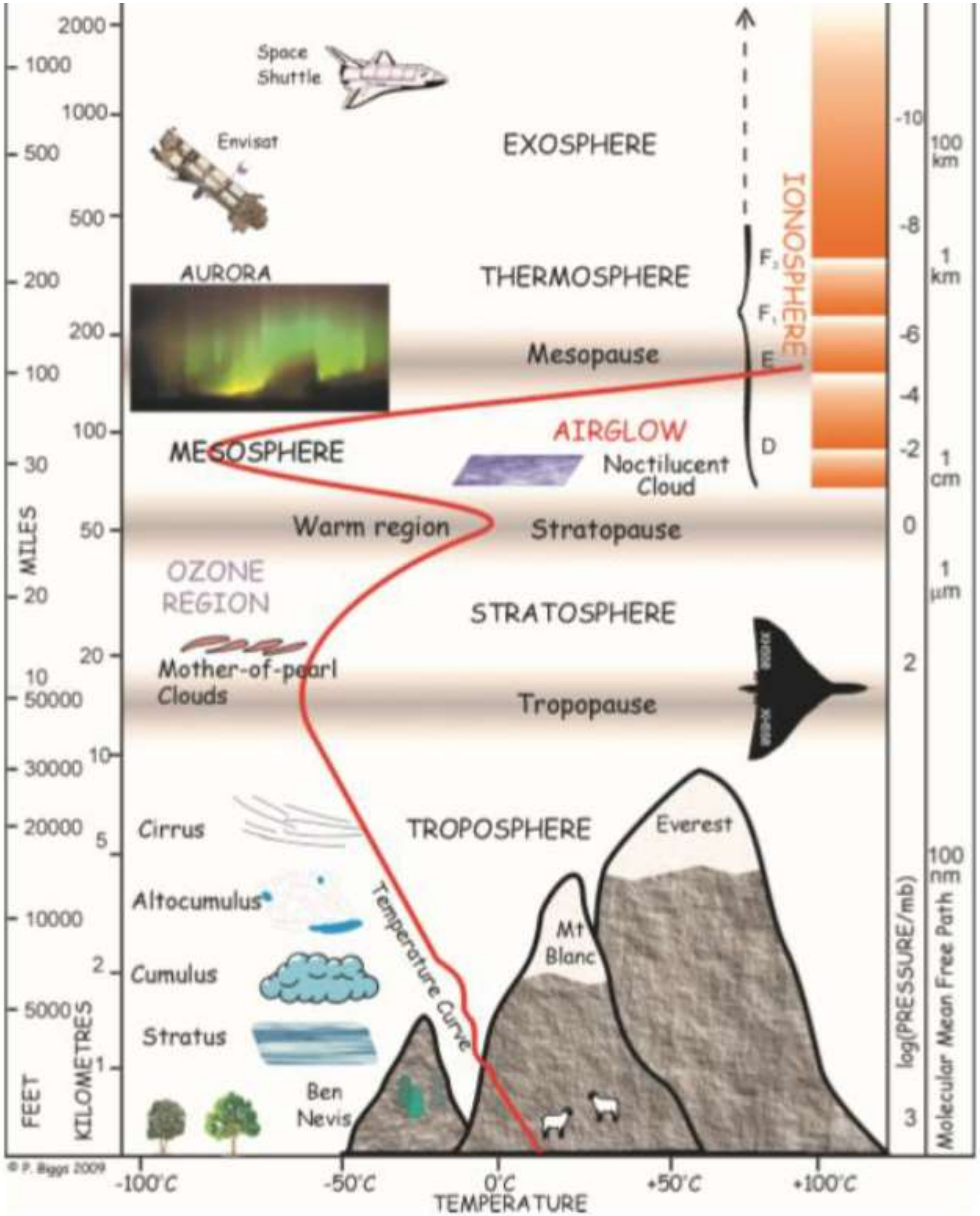


Stratosphere

strato = layered

dry

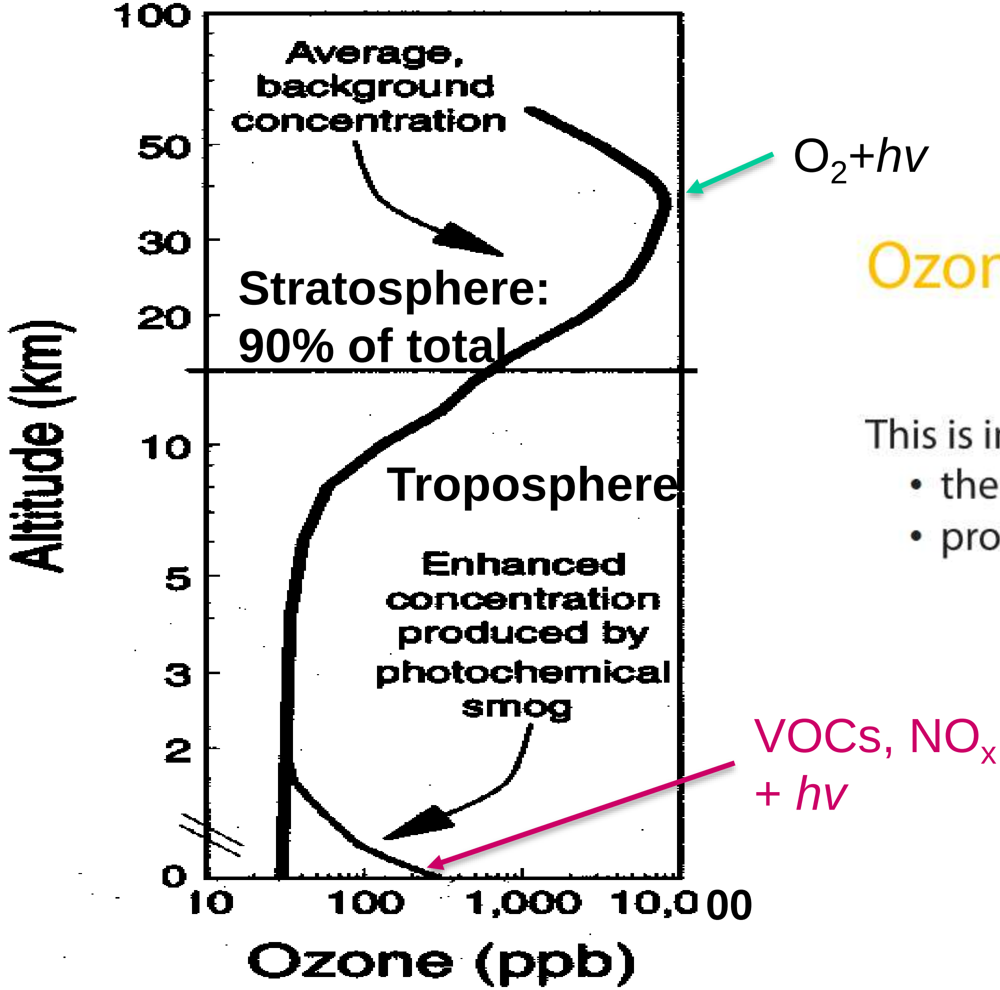
ozone rich

radiation

temperature increases with height

The stratospheric ozone hole

US EPA (1970):
ozone is "good up high, bad nearby"



Ozone absorbs high energy ultra-violet

This is important for:

- the thermal layering of the atmosphere
- protecting life on the Earth surface

A brief history of ozone discovery

1839: Schonbein discovers a smell from electrolysis of water, calls it ozone (Greek: to smell)

1865: Soret establishes that ozone is O_3

1913: Fabry discovers atmospheric ozone at high altitude by observing solar absorption spectrum

1927: Gotz infers vertical distribution of ozone by observing solar radiation near the horizon: peak at 25 km

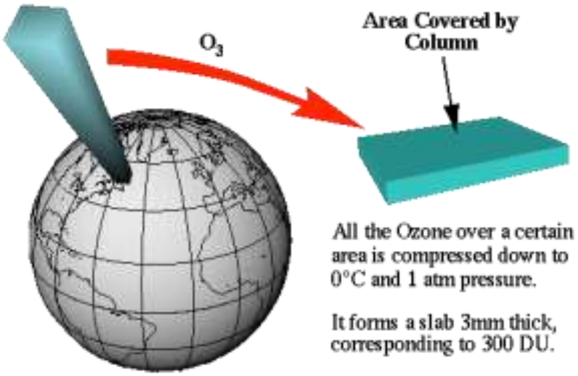
1930: Dobson maps global distribution of ozone in stratosphere

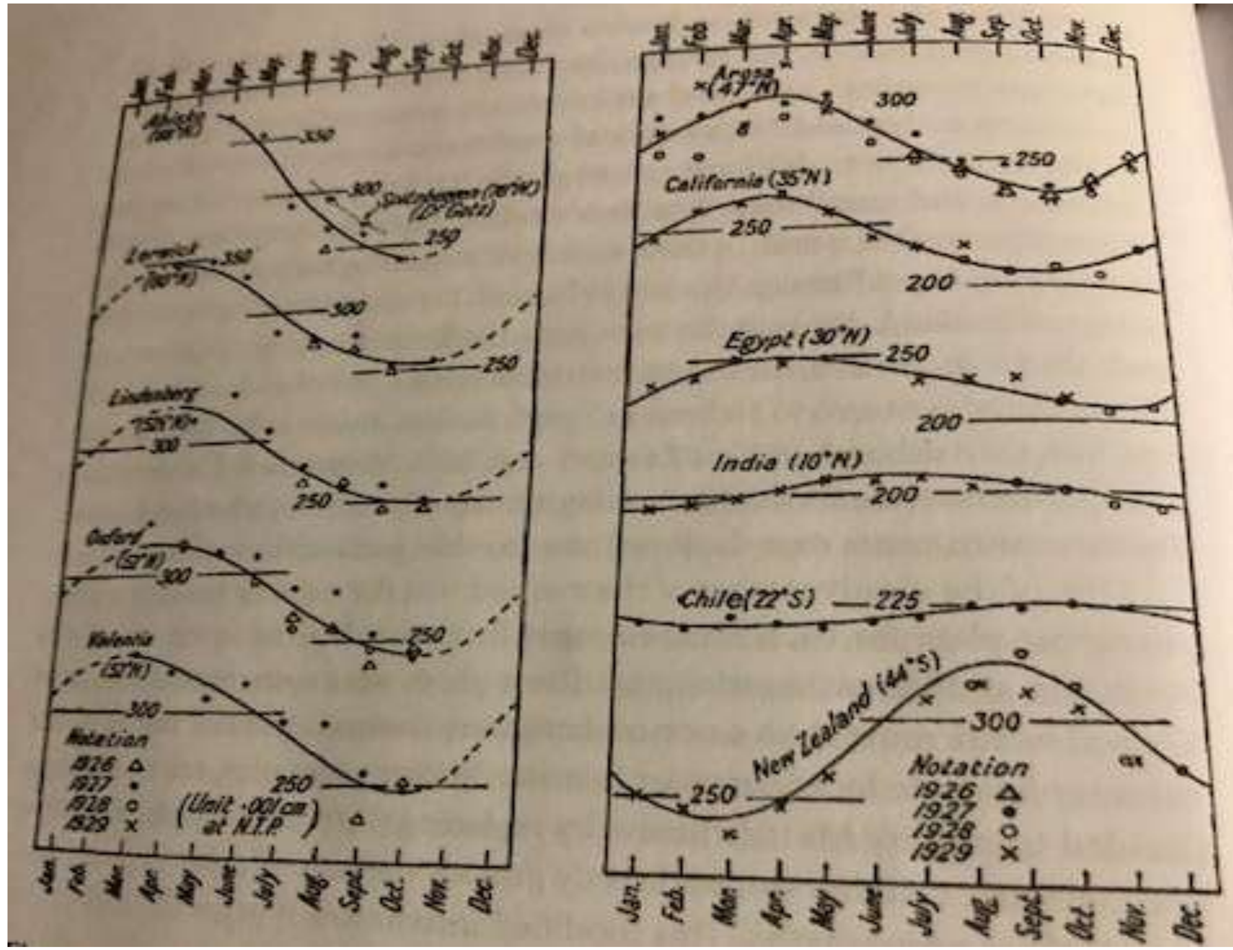
1931: Piccard does first manned ascent into stratosphere

1934: Regener measures ozone vertical profile up to 34 km from hydrogen balloon

Dobson's observations (1930)

Ozone columns reported in units of 0.01 mm pure ozone at 1 atm and 0°C (STP), now known as Dobson unit (DU): 1 DU = 2.69×10^{16} molecules cm^{-2}

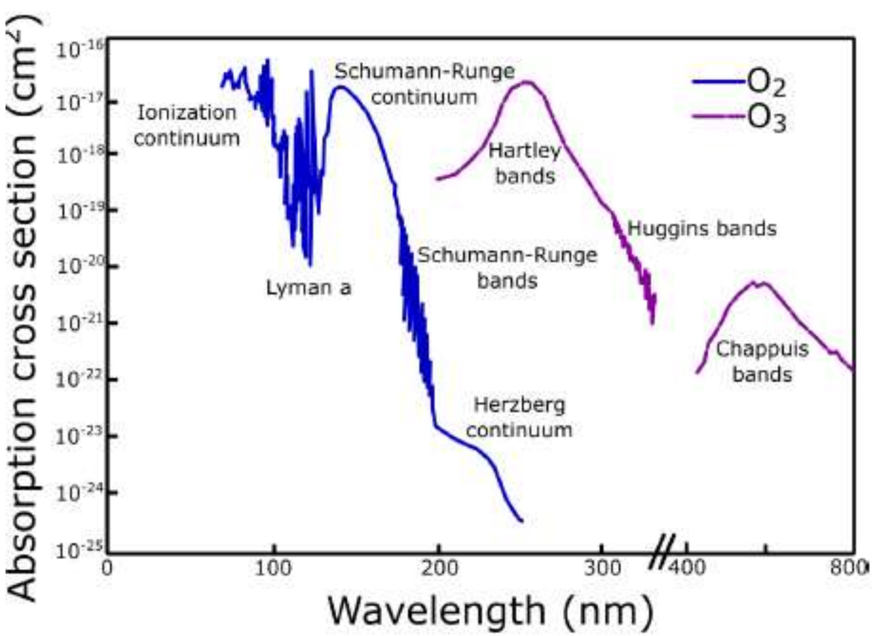




Data show thickest ozone columns at high latitudes in spring, thinnest in tropics

Stratospheric ozone

- Ozone is the most important trace constituent of the stratosphere.
- 90% of the atmosphere's ozone is found in the stratosphere – where the ozone layer forms.
- The absorption of solar radiation by ozone (and oxygen) heats the stratosphere.



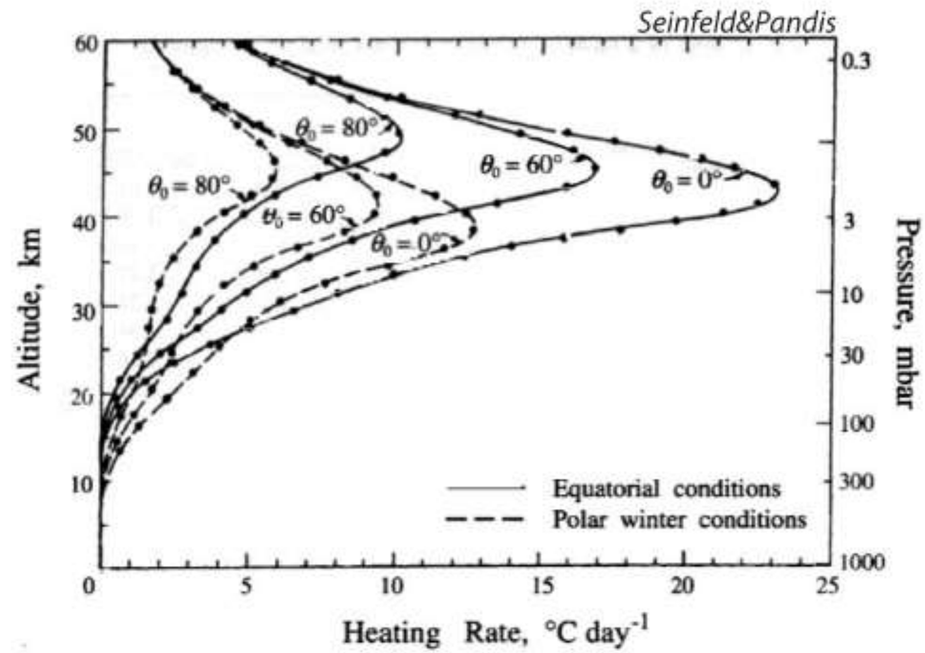
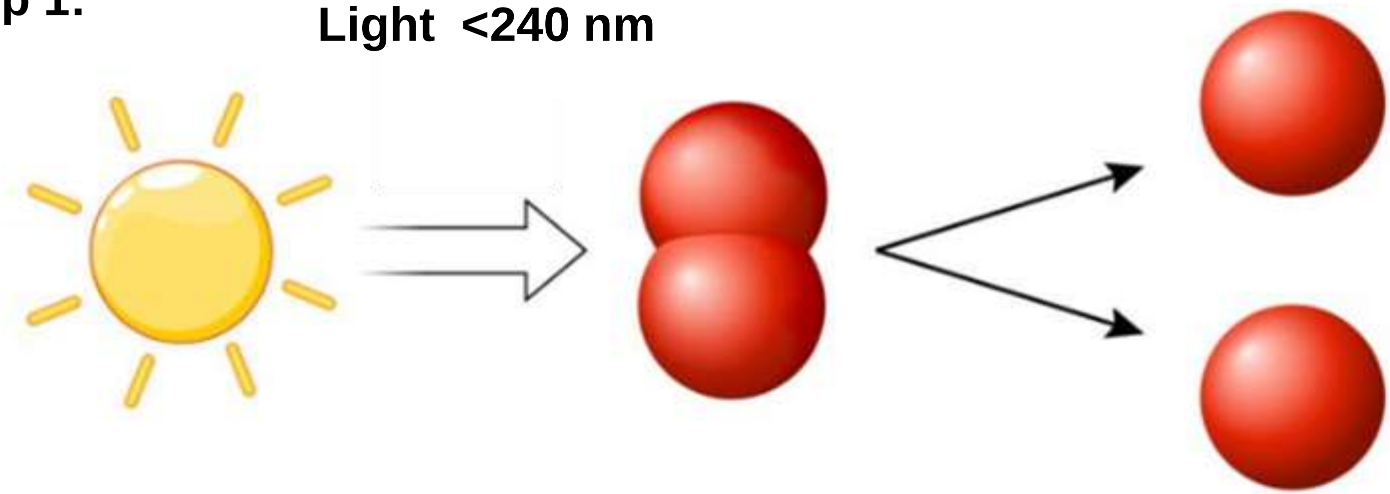


FIGURE 3.13 Heating rate from absorption of solar radiation by ozone for solar zenith angles $\theta_0 = 0°, 60°, 80°$. The solid curve is for an ozone distribution representative of equatorial conditions; the dashed curve is representative of polar winter conditions. This calculation accounts for the change in ozone mixing ratio with altitude, whereas the analytical formula (3.52) assumes mixing ra-

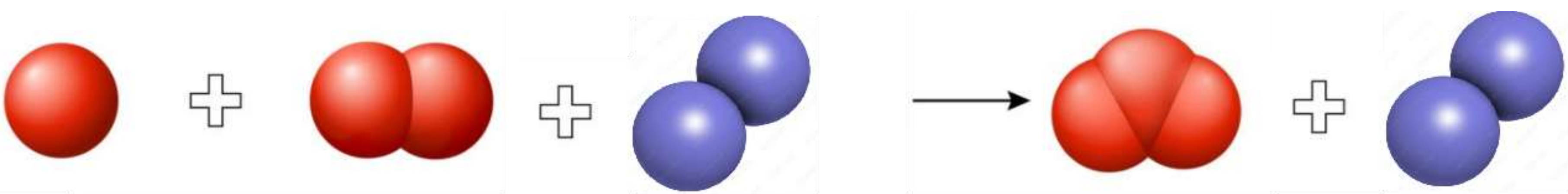
Chapman-Mechanism: Ozone formation

Step 1:

Photolysis of O_2 above 30 km
Light <240 nm



Step 2:



formation

$$O_2 + h\nu \rightarrow O + O$$

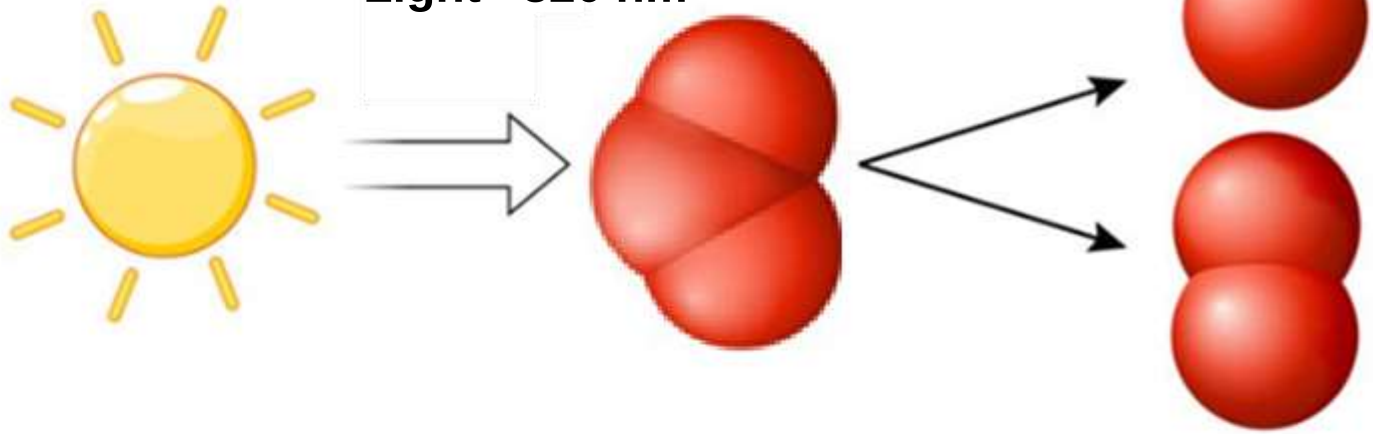
$$O + O_2 + M \rightarrow O_3 + M$$

O_3

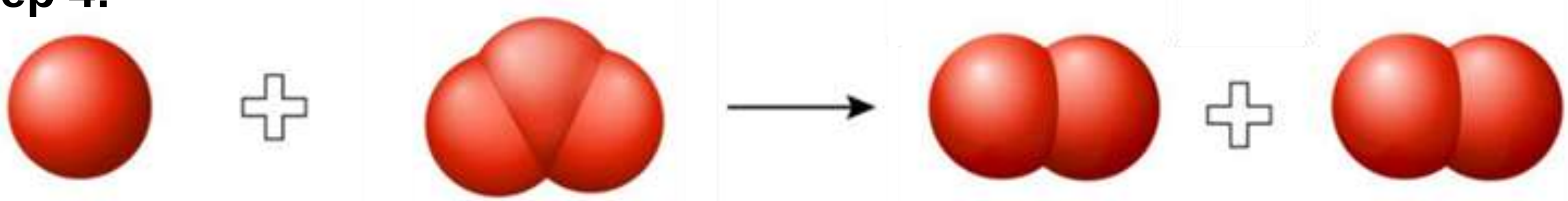
Chapman-Mechanism: Ozone destruction

Step 3:

Photolysis von O_3
Light <320 nm



Step 4:



O_3

destruction

$$O_3 + h\nu \rightarrow O + O_2$$
$$O_3 + O \rightarrow O_2 + O_2$$

How the Ozone layer protects us from UV radiation:

formation

$$\boxed{O_2 + h\nu \rightarrow O + O}$$

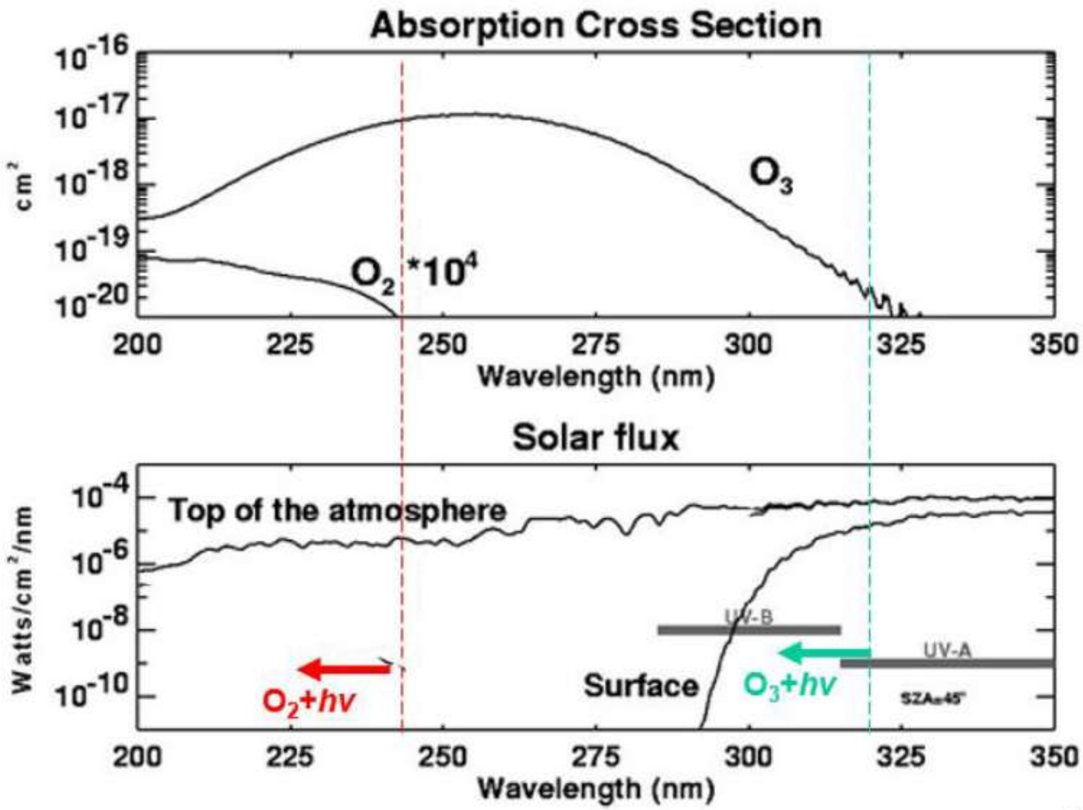
$$O + O_2 + M \rightarrow O_3 + M$$

O_3

destruction

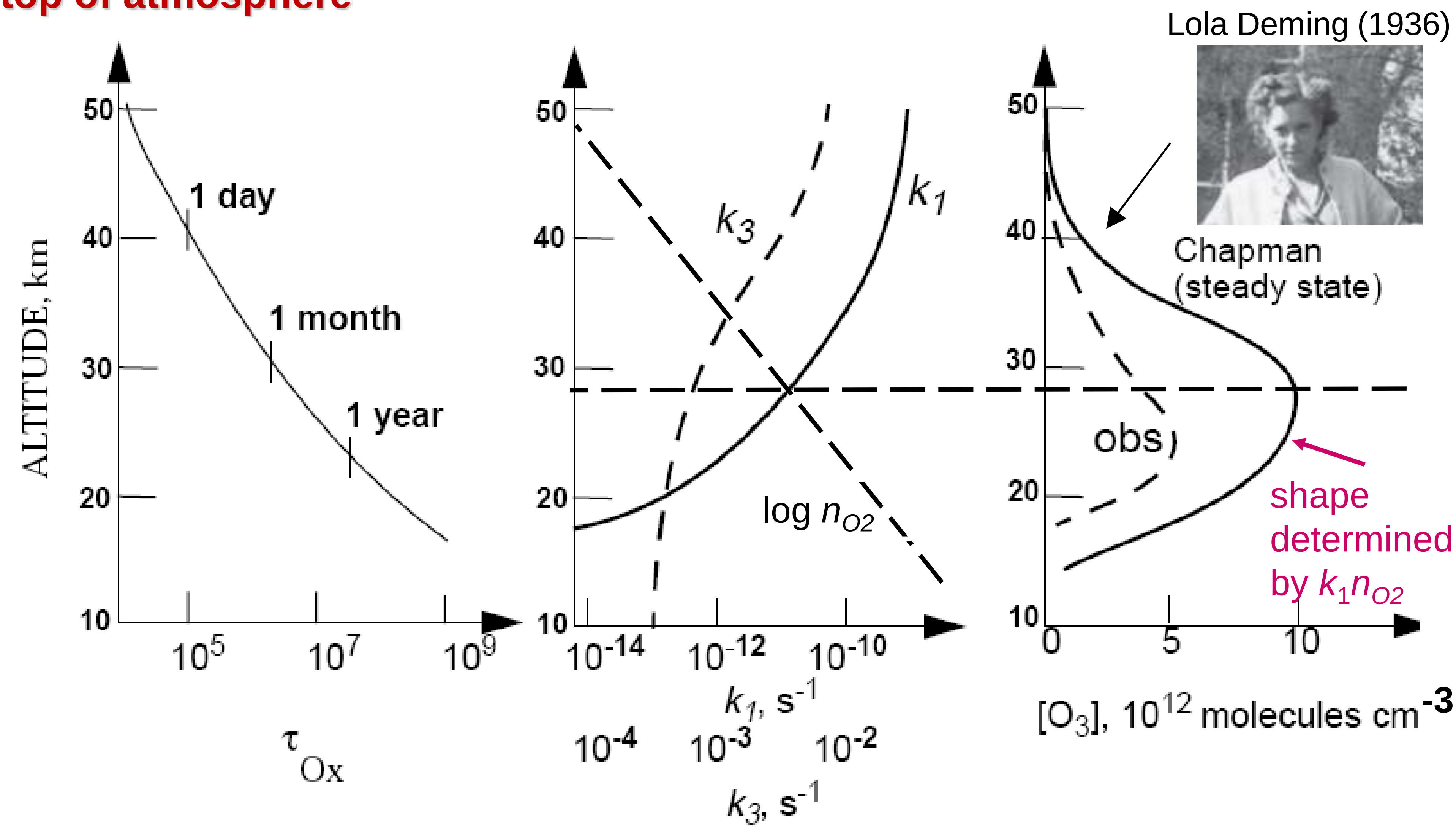
$$\boxed{O_3 + h\nu \rightarrow O + O_2}$$

$$O_3 + O \rightarrow O_2 + O_2$$



Jacob

Strong attenuation of photolysis rate constants k_1 and k_3 by ozone overhead; steady-state [O_3] must be computed numerically starting from top of atmosphere



Chapman mechanism reproduces shape, but is too high by factor 2-3 ⇨ missing sink!

Catalytic ozone destruction

1930: Chapman mechanism "oxygen only chemistry"
1940ies: Catalytic cylces destroying ozone

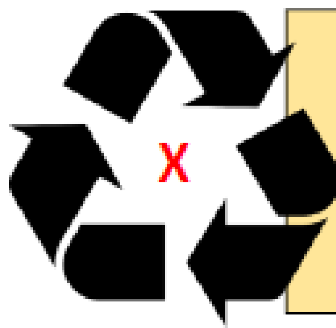
formation

$$O_2 + h\nu \rightarrow O + O$$
$$O + O_2 + M \rightarrow O_3 + M$$

O_3

destruction

$$O_3 + h\nu \rightarrow O + O_2$$
$$O_3 + O \rightarrow O_2 + O_2$$



destruction

$$O_3 + X \rightarrow XO + O_2$$
$$XO + O \rightarrow X + O_2$$

Natural cycles: HOx as a catalyst

Initiation

$$O_3 + h\nu \rightarrow O_2 + O(^1D)$$

$$H_2O + O(^1D) \rightarrow 2OH \qquad (90\%)$$

$$CH_4 + O(^1D) \rightarrow OH + CH_3 \qquad (10\%)$$

Propagation

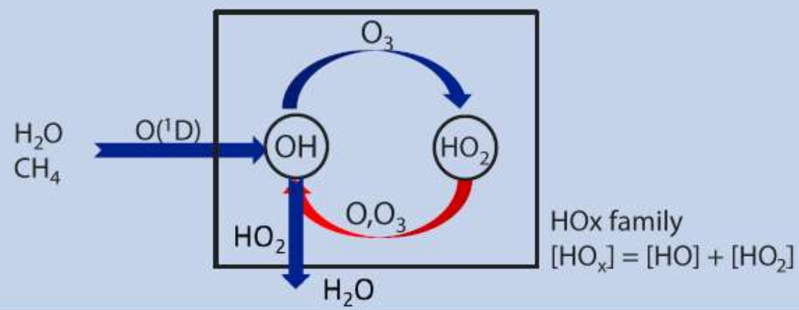
$$OH + O_3 \rightarrow HO_2 + O_2$$

$$HO_2 + O \rightarrow HO + O_2$$

or $HO_2 + O_3 \rightarrow HO + 2O_2$

$$O_3 + O \rightarrow 2O_2$$

or $2O_3 \rightarrow 3O_2$

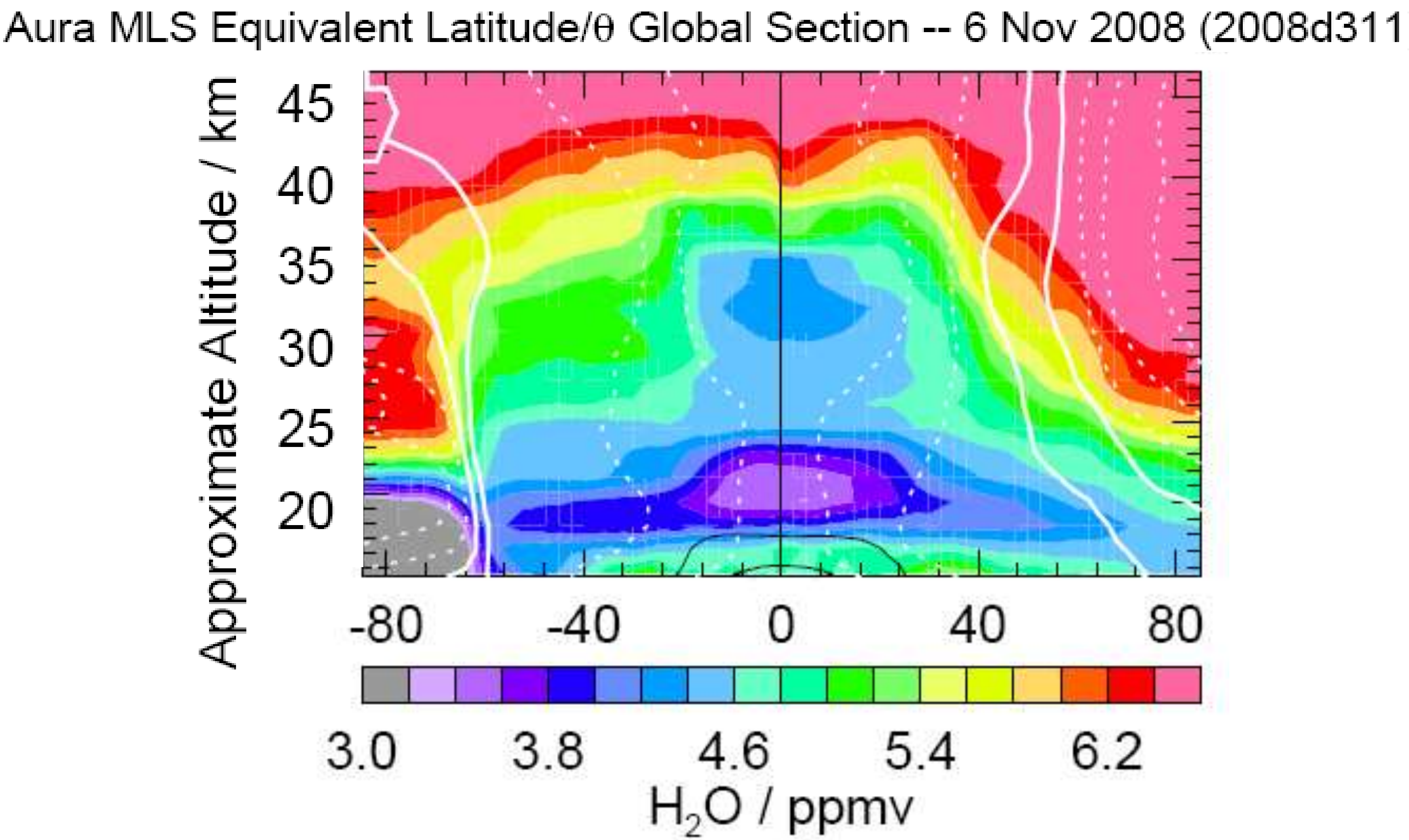


Termination

$$OH + HO_2 \rightarrow H_2O + O_2$$

NO_2 + OH + M → $\mathbf{HNO_3}$ + M

Where does the water come from?



Source of water in the stratosphere: transport from troposphere

NOx as catalyst

Initiation:

$$N_2O + h\nu \rightarrow N_2 + O(^1D).(R1)$$
$$N_2O + O(^1D) \rightarrow N_2 + O_2(R2)$$
$$N_2O + O(^1D) \rightarrow 2\,NO(R3)$$

R3 only 10%

Do nothing cycle is the rule:

$$NO + O_3 \rightarrow NO_2 + O_2$$
$$NO_2 + h\nu \rightarrow NO + O$$
$$O + O_2 + M \rightarrow O_3 + M$$

Propagation:
Happens to a small percentage of the NO_2

$$NO + O_3 \rightarrow NO_2 + O_2$$
$$NO_2 + O \rightarrow NO + O_2$$
$$\overline{O_3 + O \rightarrow 2O_2}$$

Termination (day):

$$HNO_3 + h\nu \rightarrow NO_2 + OH$$

If the HNO_3 forms ice-clouds the particles settle out of the stratosphere

Termination (night):

$$NO_2 + O_3 \rightarrow NO_3 + O_2$$
$$NO_3 + NO_2 + M \rightarrow N_2O_5 + M$$

Reactivation:

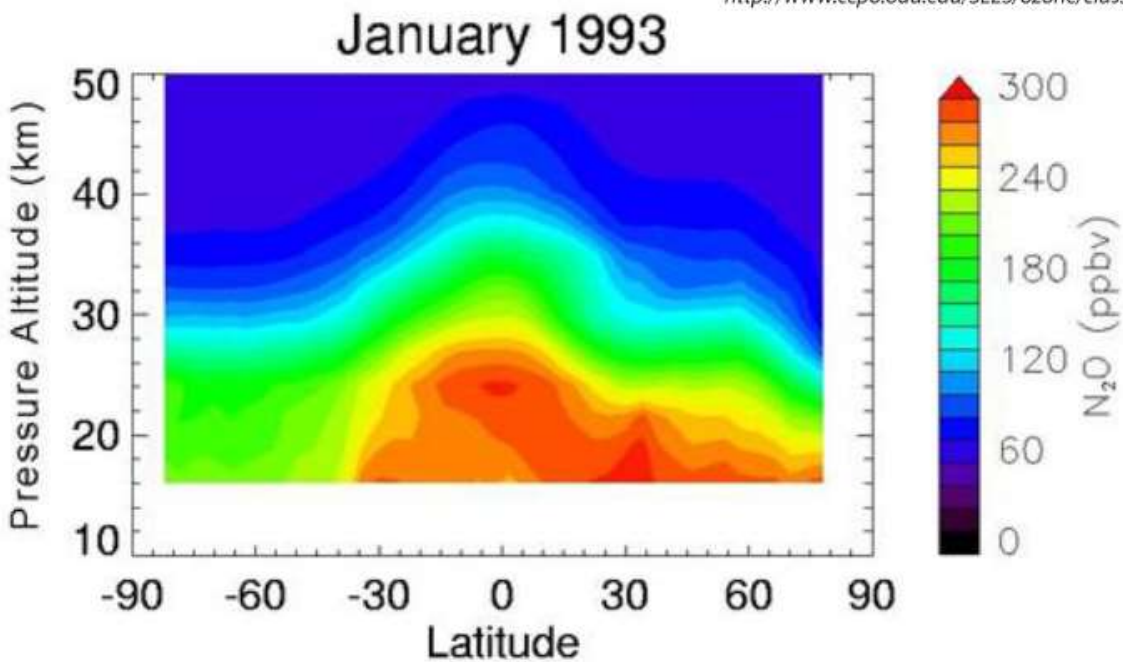
$$HNO_3 + h\nu \rightarrow NO_2 + OH$$
$$N_2O_5 + h\nu \rightarrow NO_3 + NO_2$$
$$NO_3 + h\nu \rightarrow NO_2 + O$$

How does N_2O get into the stratosphere?

Typical N_2O distribution in the stratosphere as seen by CLEAS, January 1993.

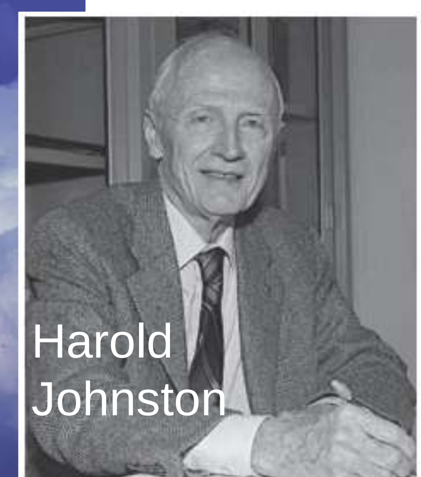
http://www.ccpo.odu.edu/SEES/ozone/class/Chap_5/5_Js/5-16.jpg

Reducing fertilizer overuse and improving nitrogen management is a form of climate change mitigation that protects the ozone layer!



- main influx of N_2O in tropics, slowly advecting upwards
- the higher N_2O reaches, the more gets photolyzed by more energetic photons

Supersonic stratospheric aircraft (SSTs) and their effect on ozone (Johnston, 1971)



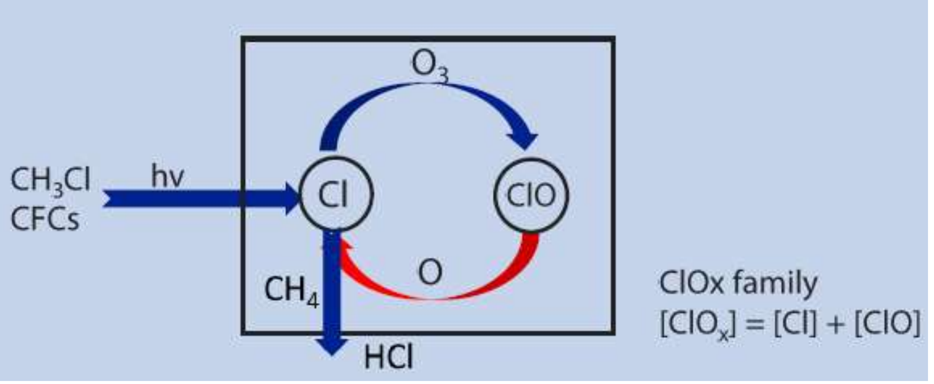
ClOx as catalyst

Initiation:

$$CH_3Cl + h\nu \rightarrow Cl + CH_3 \text{ (natural)}$$
$$CFCl_3 + h\nu \rightarrow CFCl_2 + Cl \text{ (anthropogenic)}$$

Propagation:

$$Cl + O_3 \rightarrow ClO + O_2$$
$$ClO + O \rightarrow Cl + O_2$$
$$\overline{}$$
$$O_3 + O \rightarrow 2O_2$$



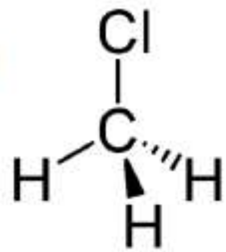
Termination:

$$Cl + CH_4 \rightarrow HCl + CH_3$$

$$ClO + NO_2 \rightarrow ClONO_2$$

Where does the Cl come from?

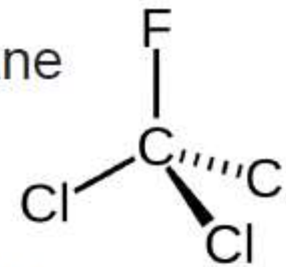
Methylchloride (CH_3Cl)



biogenic sources

anthropogenic sources

Trifluorochloromethane



…and many others

Chlorofluorocarbons (CFCs)

anthropogenic sources

Ozone chemistry

1930: Chapman mechanism "oxygen only chemistry"
1940ies: Catalytic cylces destroying ozone
1974: Chlorine catalytic cycle
1974: Chlorofluoromethanes destroy ozone
1977: US government starts acting to ban CFCs

Stratospheric sink for chlorofluoromethanes : chlorine atomc-atalysed destruction of ozone

Mario J. Molina & F. S. Rowland

Department of Chemistry, University of California, Irvine, California 92664

Chlorofluoromethanes are being added to the environment in steadily increasing amounts. These compounds are chemically inert and may remain in the atmosphere for 40–150 years, and concentrations can be expected to reach 10 to 30 times present levels. Photodissociation of the chlorofluoromethanes in the stratosphere produces significant amounts of chlorine atoms, and leads to the destruction of atmospheric ozone.

photolytic dissociation to $CFCl_2$ + Cl and to CF_2Cl + Cl, respectively, at altitudes of 20–40 km. Each of the reactions creates two odd-electron species—one Cl atom and one free radical. The dissociated chlorofluoromethanes can be traced to their ultimate sinks. An extensive catalytic chain reaction leading to the net destruction of O_3 and O occurs in the stratosphere:

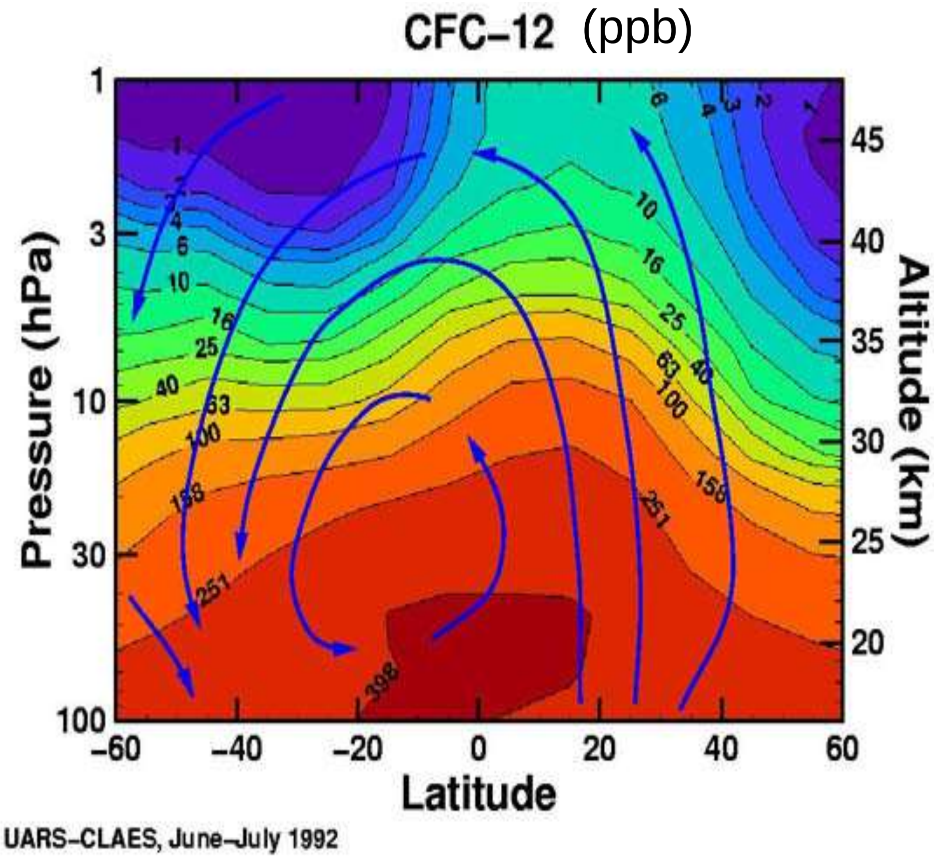
$$Cl + O_3 \rightarrow ClO + O_2 \quad (1)$$

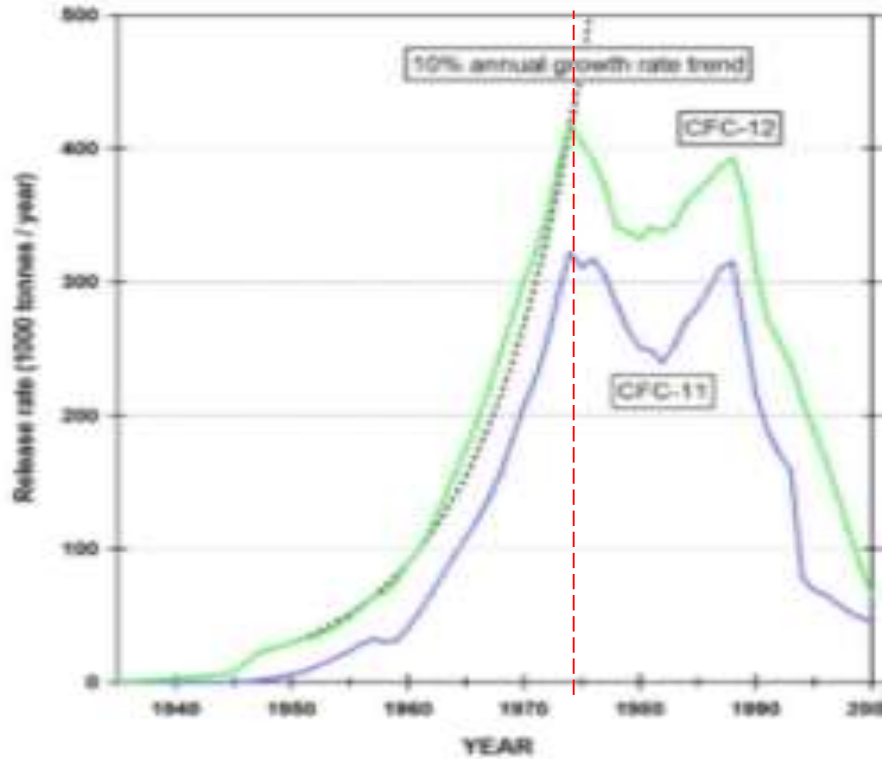
$$ClO + O \rightarrow Cl + O_2 \quad (2)$$

This has important chemical consequences. Under most conditions in the Earth's atmospheric ozone layer, (2) is the

Rapid growth of CFCs in the 1960s and 70s

aerosol spray cans





- Industrial gases valued for their inertness…but they photolyze in stratosphere
- Most important are CFC-11 ($CFCl_3$) and CFC-12 (CF_2Cl_2)
- Large-scale production started in 1950s, growth in 1970s approached 10%/year

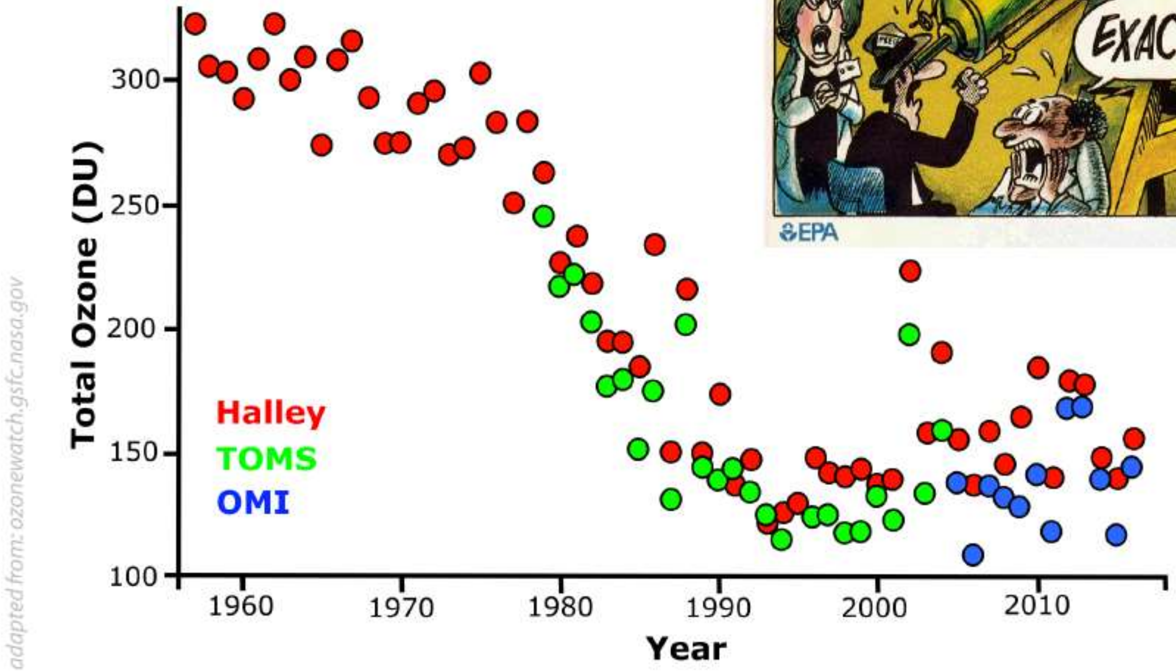
1985: ozone hole first reported by British Antarctic Survey (Farman et al., 1985)

1985: 40% missing ozone in Antarctica (Halley 1985)

- satellite observations confirm ozone loss
- start of enhanced scientific research and actions

Why was the ozone hole not spotted earlier?

Satellite instruments did not detect the ozone hole because the ground software that processed the data from those devices simply rejected readings that were either overly low or overly high



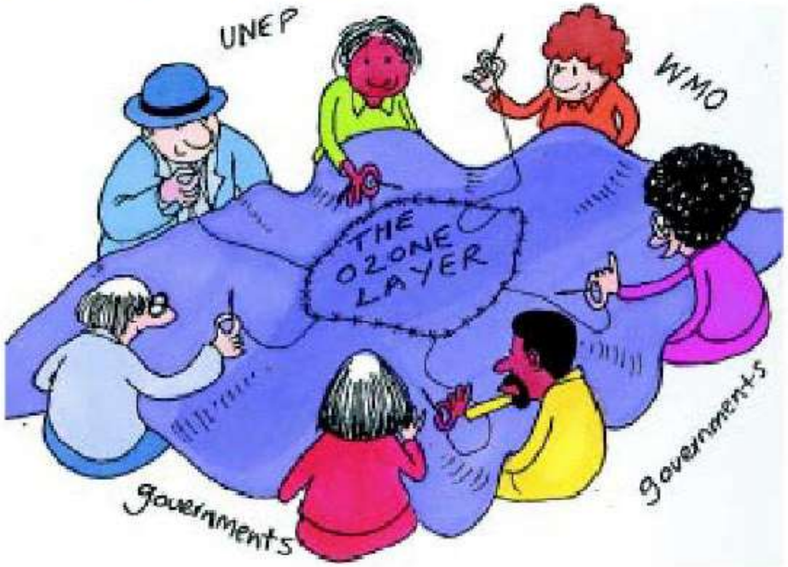
The Montreal Protocol on Substances that Deplete the Ozone Layer

unep.org/ozonaction

The Montreal Protocol

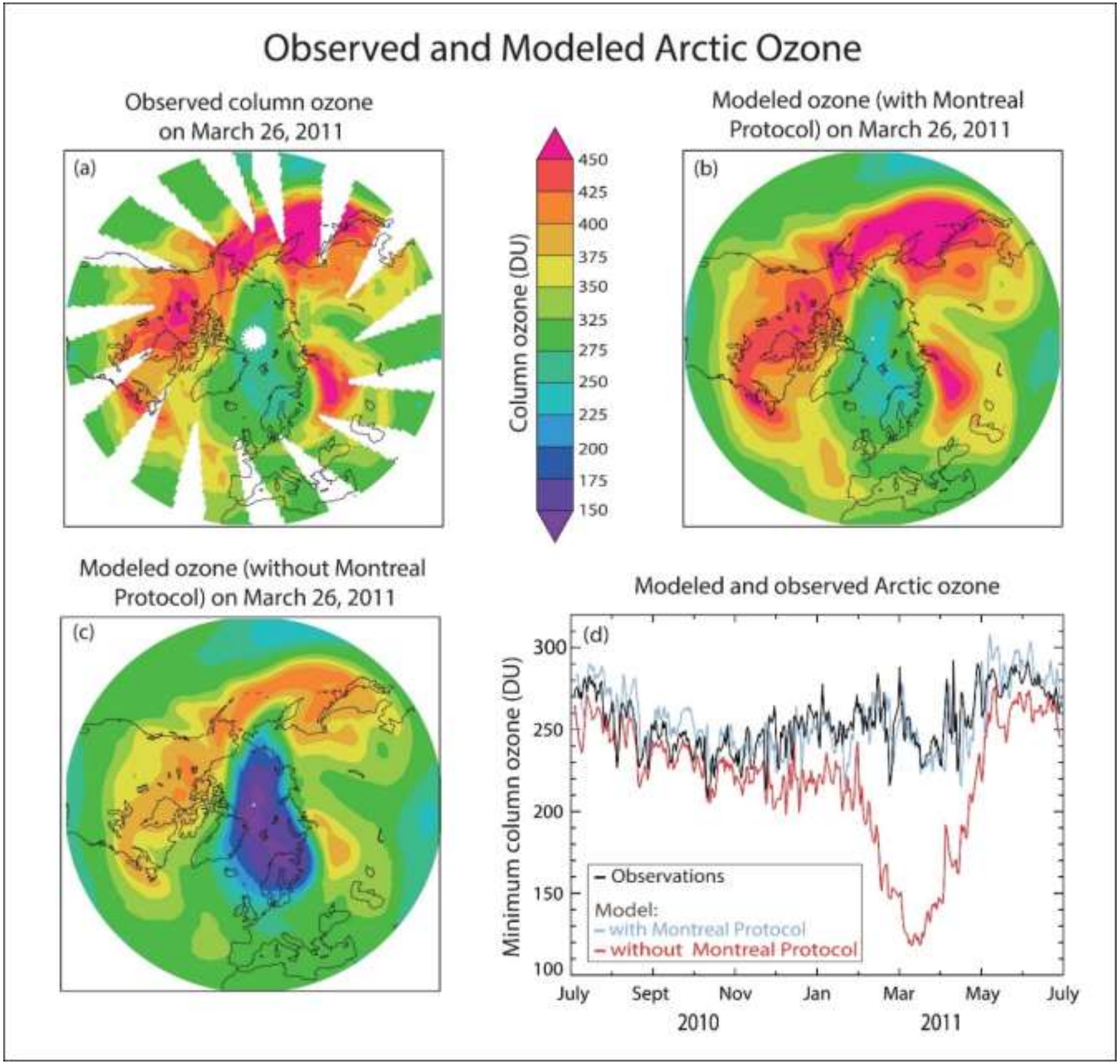
The Montreal Protocol on Substances that Deplete the Ozone Layer is the landmark multilateral environmental agreement that regulates the production and consumption of nearly 100 man-made chemicals referred to as ozone depleting substances (ODS). When released to the atmosphere, those chemicals damage the stratospheric ozone layer, Earth's protective shield that protects humans and the environment from harmful levels of ultraviolet radiation from the sun. Adopted on 15 September 1987, the Protocol is to date the only UN treaty ever that has been ratified every country on Earth - all 198 UN Member States.

1981: UNEP starts negotiations
1987: Montreal Protocol
1990: London Amendment → completely phase out CFCs by 2020 & Fund
1992: Copenhagen Amendment → phase out CFC by 1996 & HCFCs by 2030
...
2016: Kigali Amendment → phase of HFCs



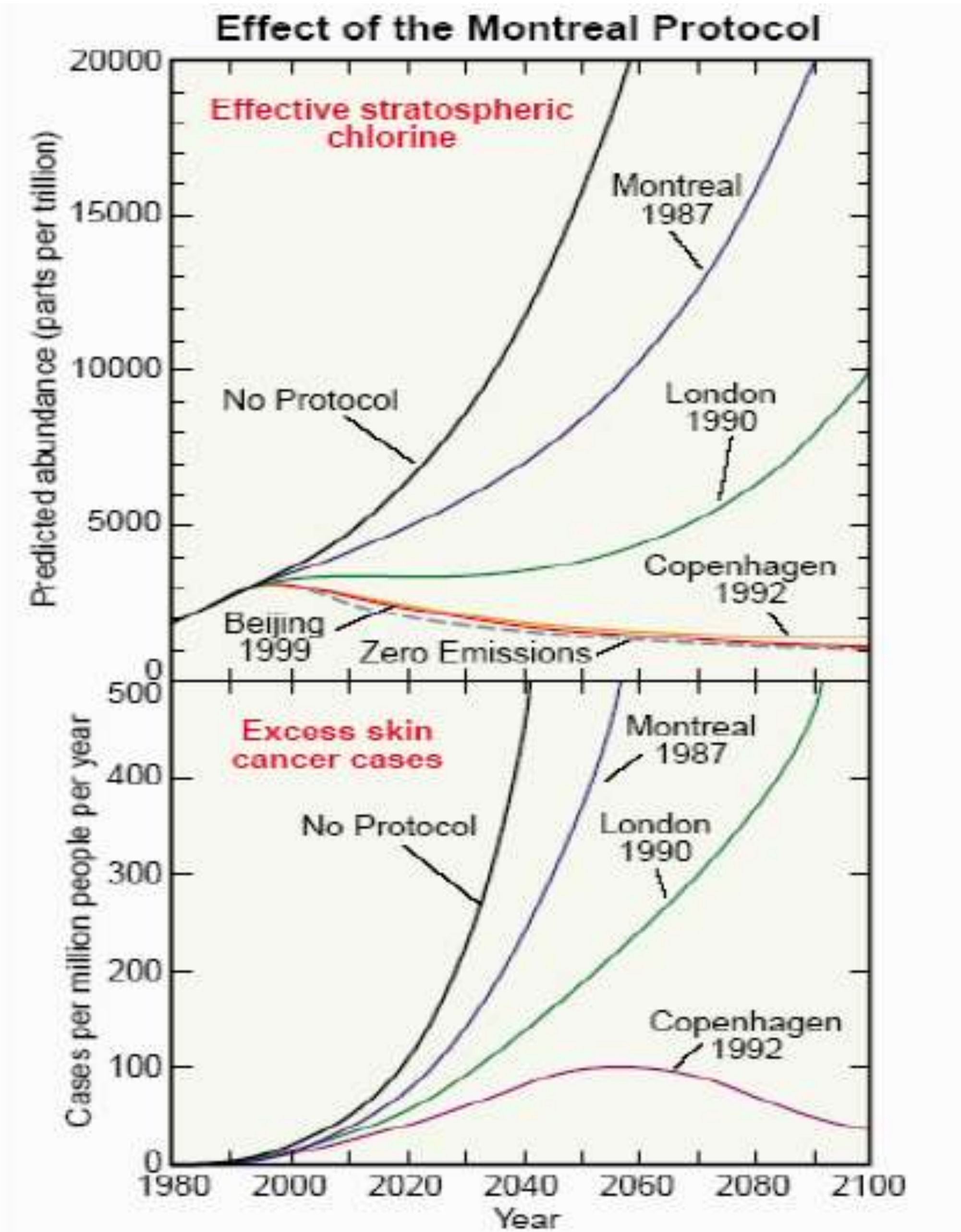
https://ozone.unep.org/sites/default/files/2019-08/ozone-cartoons.pdf

Montreal Protocol avoided the Arctic ozone hole

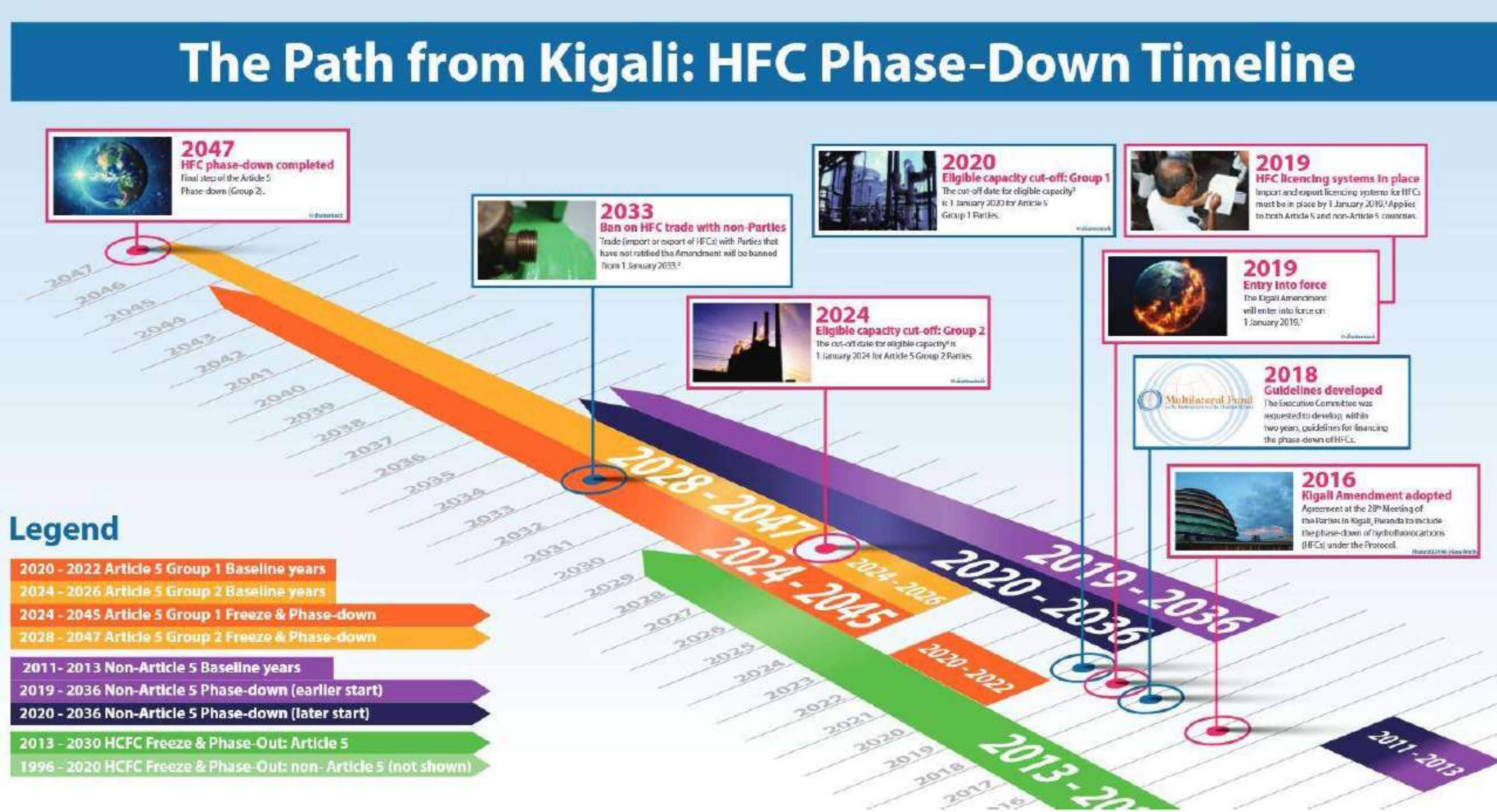


WMO [2018]

Montreal Protocol avoided many deaths

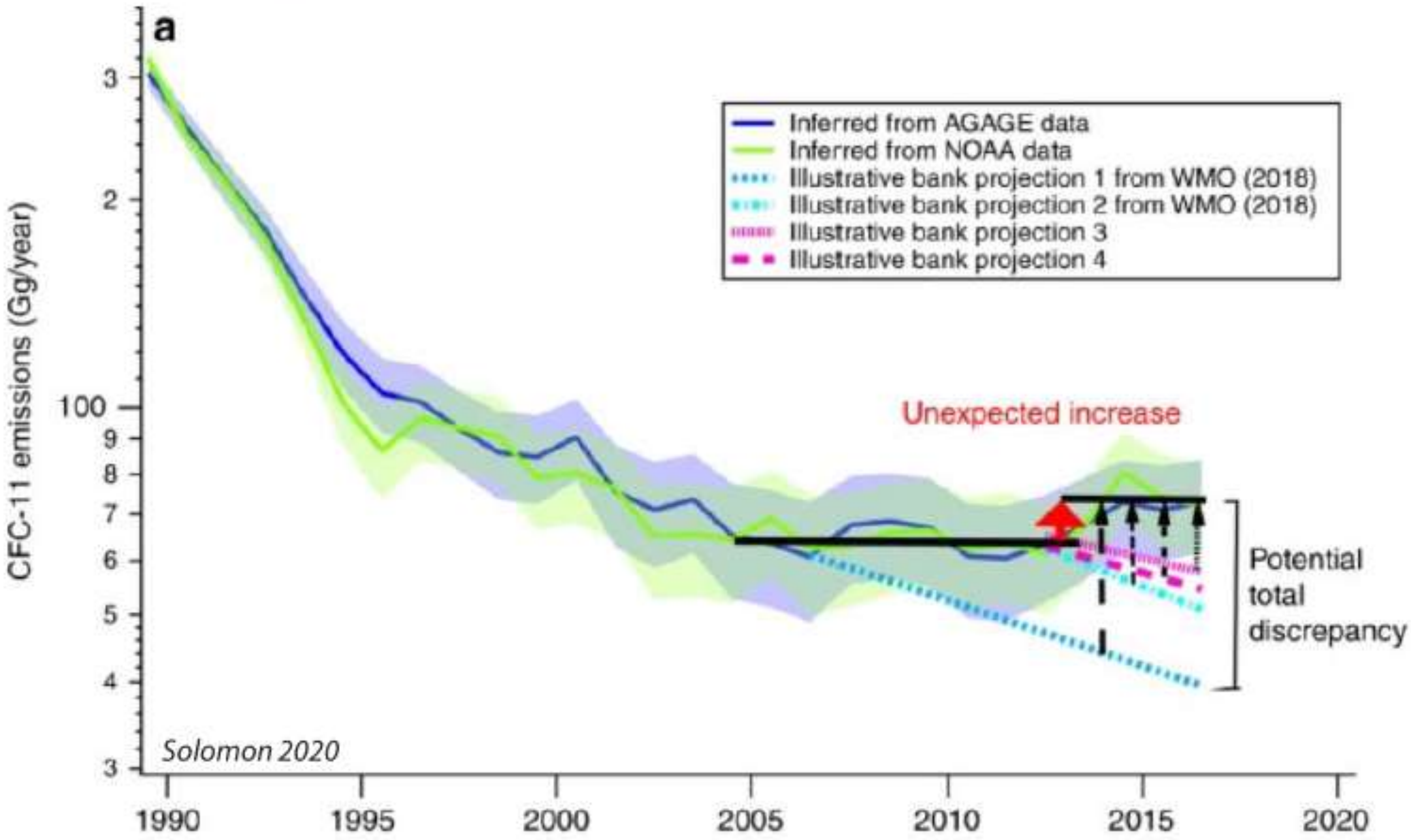


Kigali amendment

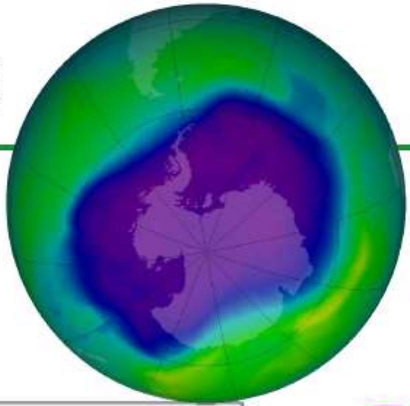


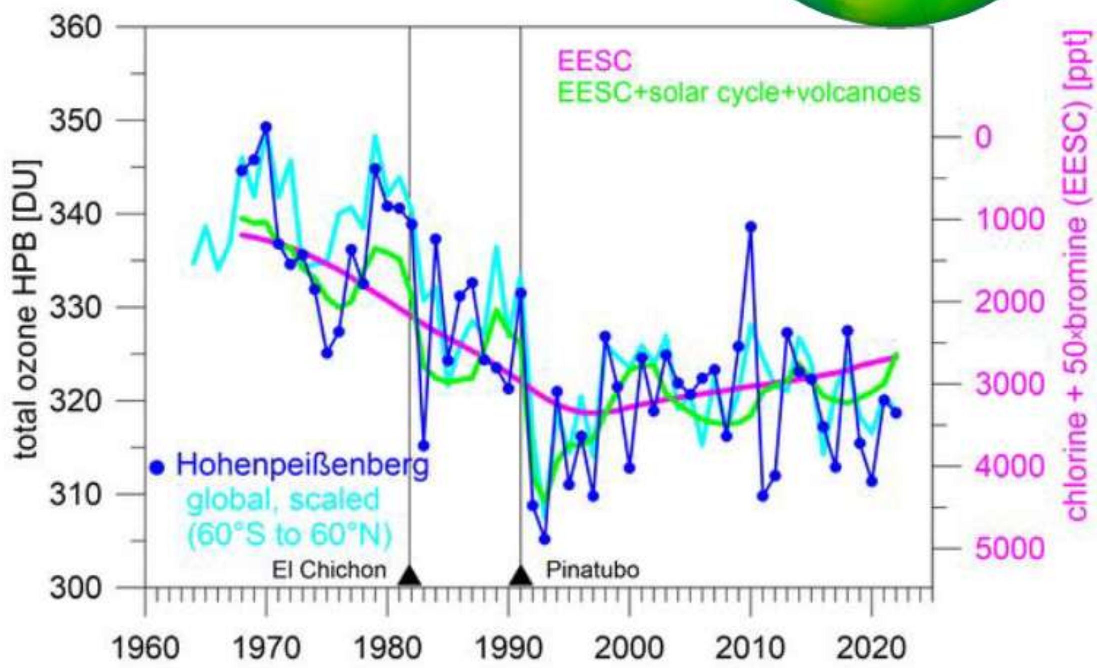
Is the Montreal Protocol successful?

Fig. 1: Emissions of CFC-11



The same trends can be observed at Hohenpeißenberg

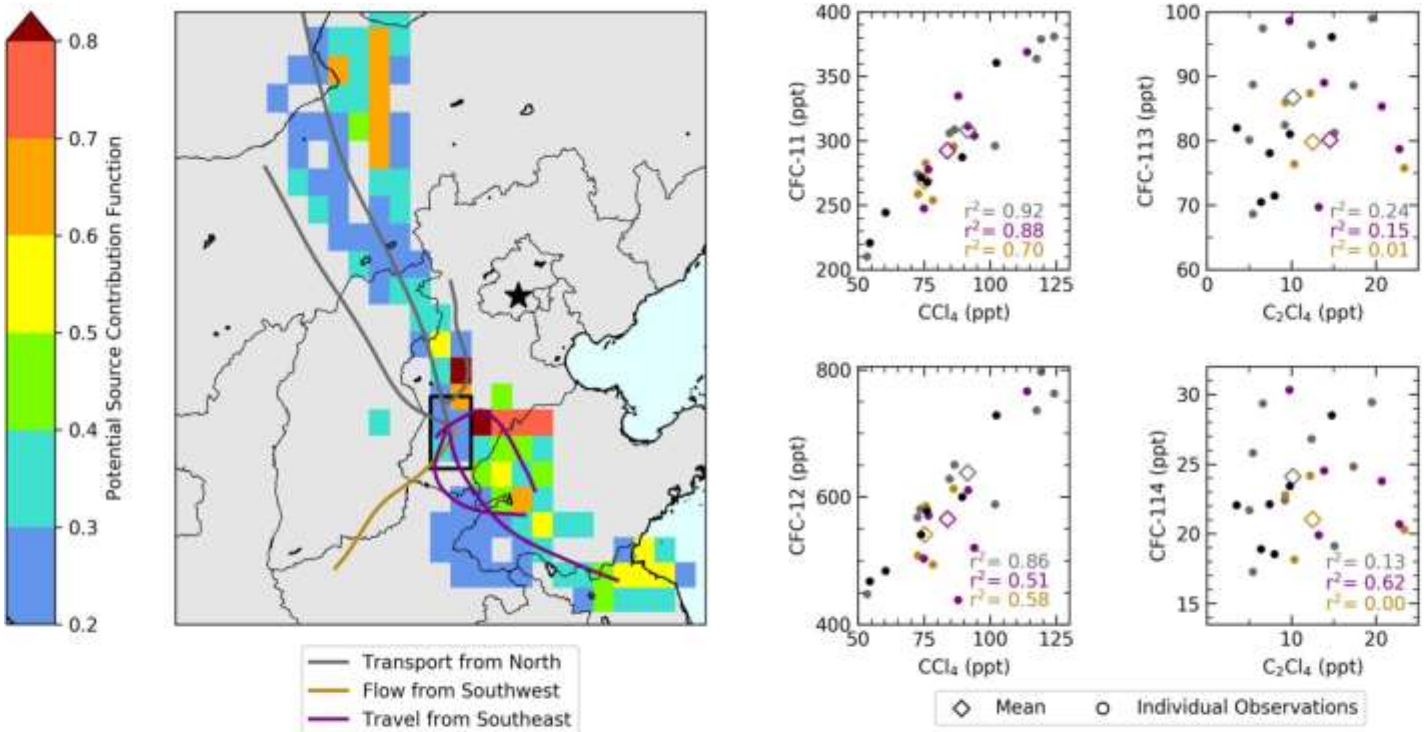




DWD, German Weather Services

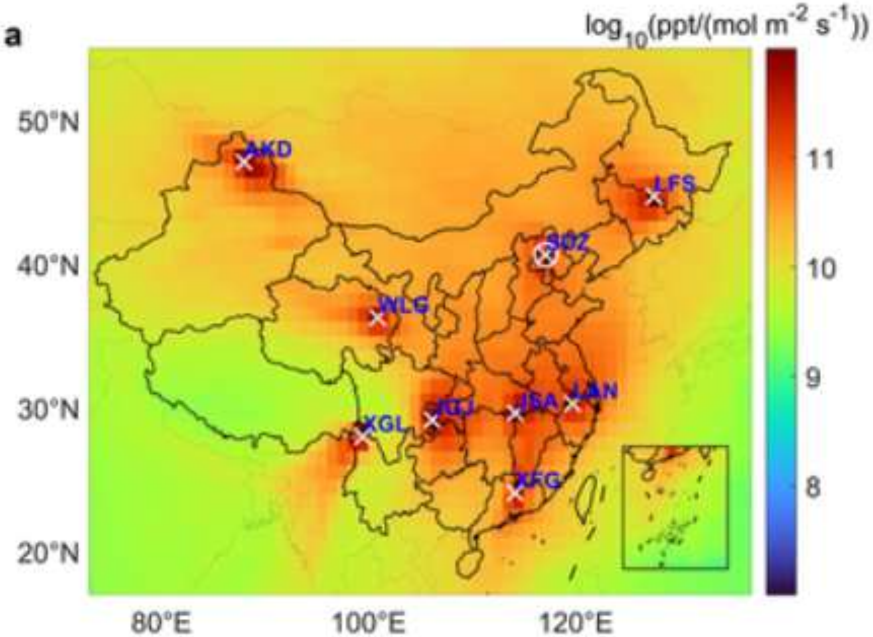
Illicit manufacturing activities

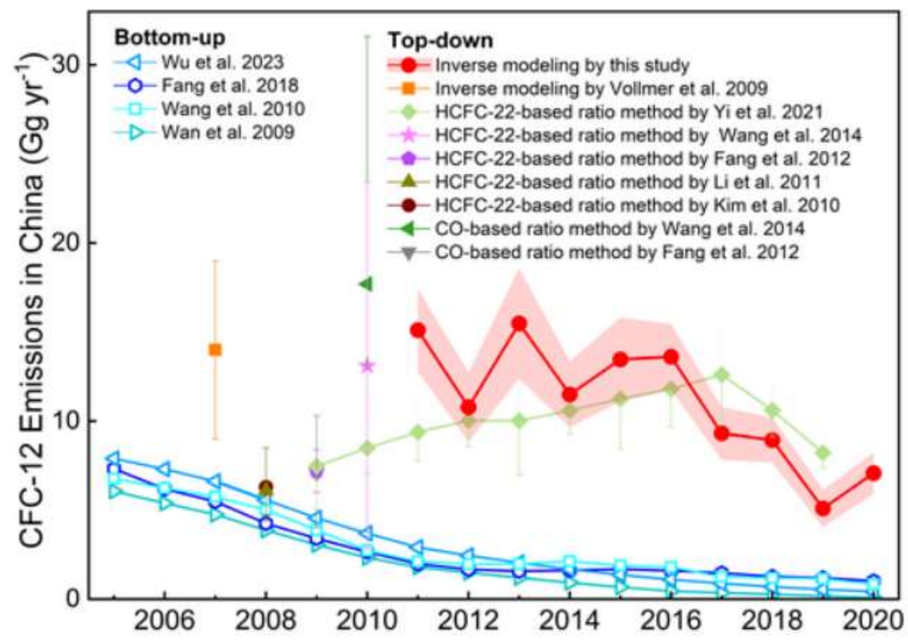
Airborne Observations of CFCs Over Hebei Province, China in Spring 2016



JGR Atmospheres, Volume: 126, Issue: 18, First published: 29 August 2021, DOI: (10.1029/2021JD035152)

Illicit manufacturing activities



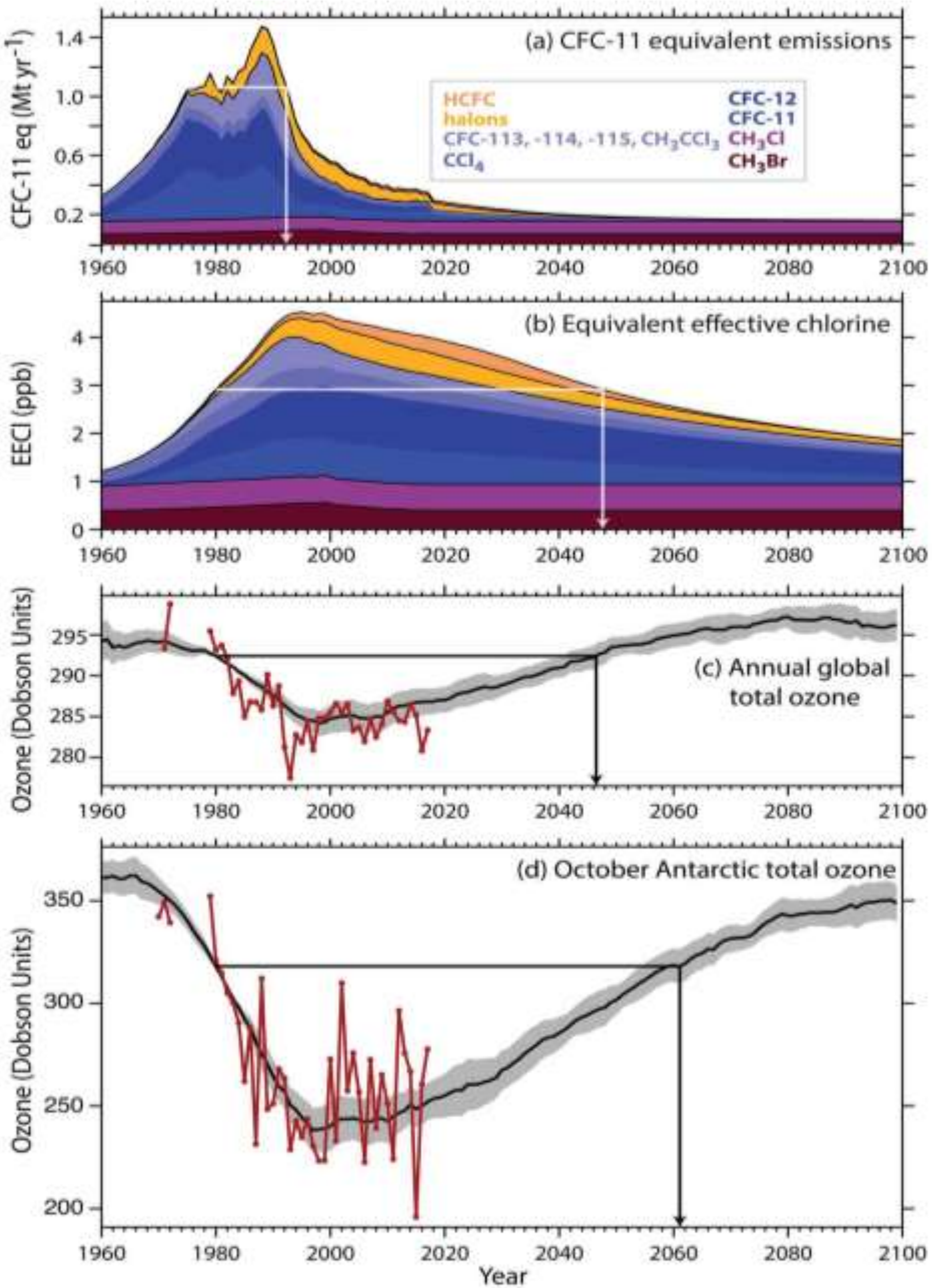


NEWS | 10 February 2021

Illegal CFC emissions have stopped since scientists raised alarm

Analyses suggest that China has successfully curbed production of an ozone-depleting chemical, a win for the international treaty that protects the ozone layer.

CFCs have a lasting legacy for ozone depletion

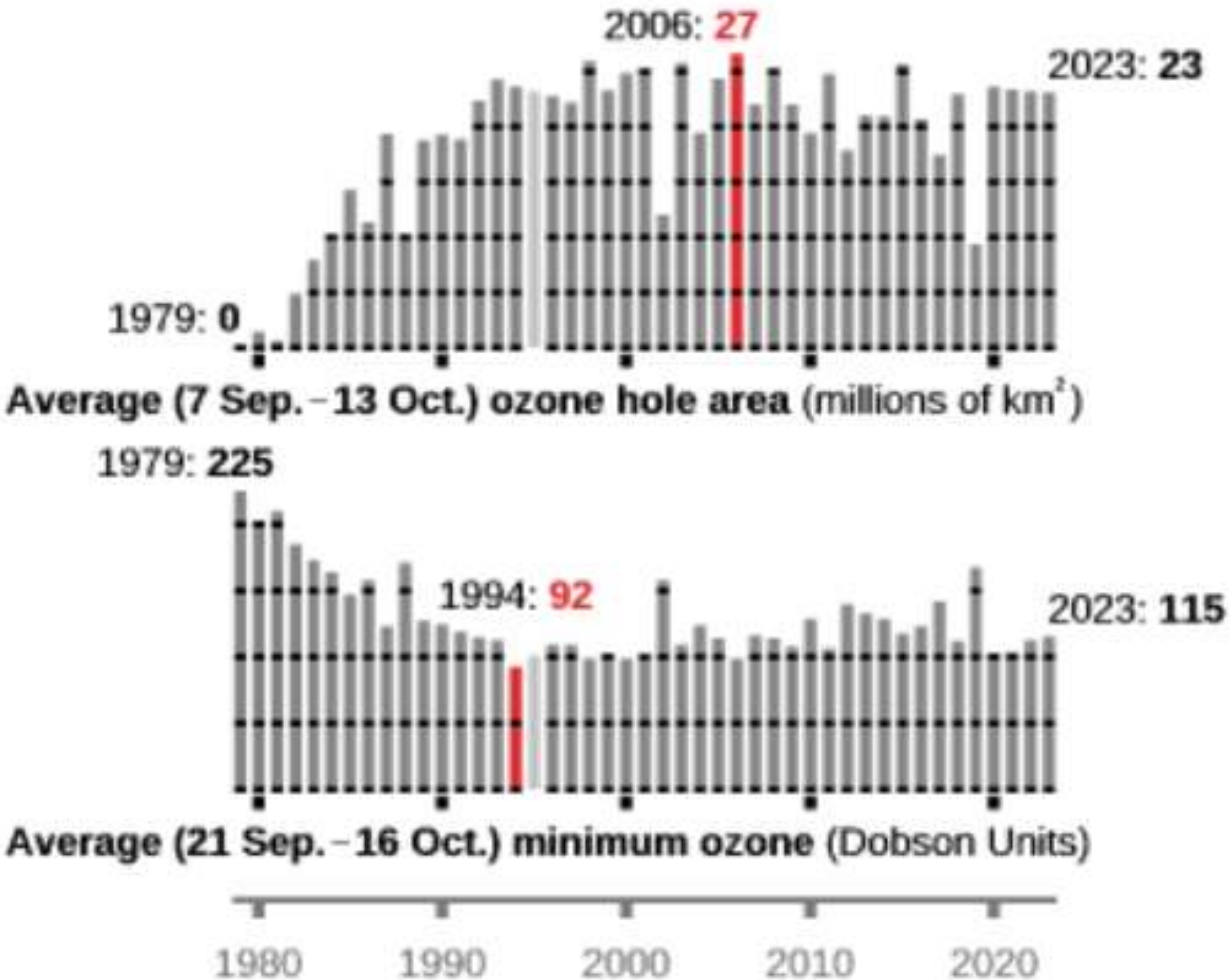


Chlorine takes a long time to decrease because of the long lifetime of CFCs…

…and therefore the recovery of the ozone layer will take the rest of this century

WMO [2018]

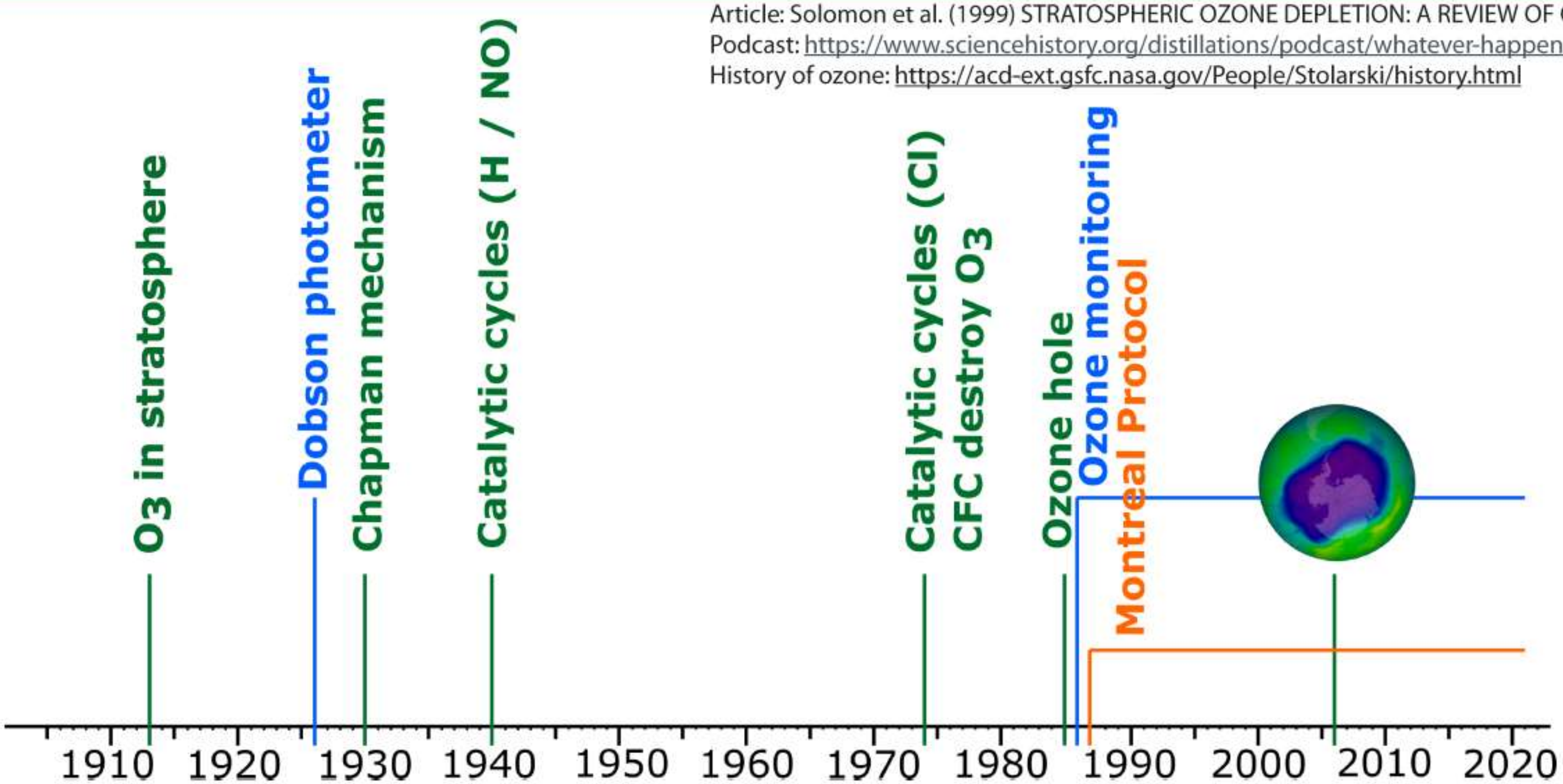
Latest data confirm that it is still too early to see Antarctic ozone recovery



http://ozonewatch.gsfc.nasa.gov/

Summary: Stratospheric ozone depletion

Article: Solomon et al. (1999) STRATOSPHERIC OZONE DEPLETION: A REVIEW OF CONCEPTS AND HISTORY
Podcast: https://www.sciencehistory.org/distillations/podcast/whatever-happened-to-the-ozone-hole
History of ozone: https://acd-ext.gsfc.nasa.gov/People/Stolarski/history.html



The adoption of the Montreal Protocol by all nations and the enforcement of the treaty even by those Nations that often act rogue shows that collective action on a global scale is possible.

We need to get to the same point on climate change. Sadly we are in a situation where new technologies have made global scale disinformation campaigns targeted a preventing collective action to foster the economic interests of a few individuals very easy.

Fun facts:

Private jet or more in business class

The most efficient climate change mitigation (long term) are scholarships for middle school girls in high fertility countries

The second most efficient climate change mitigation would be a global minimum landing fee of 1 lakh $ per touchdown on private jets in all countries

Fun facts:

EFFECT OF GROWING INCOME INEQUALITY

HOUSEHOLDS THAT EARN THIS MUCH PER YEAR	WOULD EARN THIS MUCH IF INCOME WERE STILL DISTRIBUTED LIKE IT WAS 1980	EFFECT OF GROWTH IN INEQUALITY SINCE 1980
$11,651	$14,865	-$3,214
$20,900	$25,791	-$4,891
$30,509	$36,996	-$6,487
$40,187	$48,366	-$8,179
$52,322	$61,049	-$8,727
$65,501	$74,149	-$8,648
$83,519	$89,939	-$6,420
$105,910	$108,568	-$2,658
$336,765	$239,651	+$97,114
$1,081,280	$607,362	+$473,917
$24,988,251	$7,539,104	+$17,449,147

COURTESY: POLITICSTHATWORK.COM

A wealth tax that combats income inequality is another very efficient form of climate change mitigation…

---which is why there is so much money behind sponsoring governments that represent the interest of a few rather than that of most voters.

Also remember if you are making less than 1 cr per year you are on the side of the losers rather than on the side of the winners.